

BRIAN JARRETT

Senior Backend & Platform Engineer

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PROFESSIONAL SUMMARY

Highly analytical engineering professional with a 30-year foundation in complex system validation, data integrity, and autonomous technology adaptation. Proven track record of entering unfamiliar technical ecosystems, rapidly mastering complex architectures, and becoming a Subject Matter Expert (SME) for the organization's technology stack. Expert in diagnosing legacy infrastructure, verifying data quality at the source to prevent the propagation of bad data and ensure the integrity of the final output, and architecting robust ETL pipelines that merge disparate inputs into high-integrity sources of truth. An objective, narrative-free troubleshooter who translates intricate system data into actionable executive metrics, applying pragmatic innovation—including targeted AI/LLM workflows—strictly where it maximizes operational value.

SKILLS

Data & Systems Engineering

- **Python (Pandas, SQLAlchemy):** Large-scale data manipulation, automated reporting, ETL pipelines, and API integrations.
- **ETL Pipelines:** Architecting high-integrity data pipelines, database migrations, and schema validation.
- **PostgreSQL, MySQL:** Relational database operations, replication, schema migration, and performance query tuning.
- **Validation & QA Testing:** Rigorous test case design, orchestration testing, and end-to-end verification methodologies.
- **AI & RAG Orchestration:** Local multi-agent LLM orchestration (LiteLLM, Hermes), vector databases (Qdrant), and graph analytics.

Systems & Distributed Scale

- **Cisco UCS Hardware:** Enterprise bare-metal compute and fabric interconnect administration.
- **OpenStack:** Public/private cloud orchestration, scheduler tuning, and infrastructure verification.
- **vSphere / ESXi / Proxmox:** High-density enterprise virtualization, clustering, and VM lifecycle automation.
- **Ceph Block Storage (RBD):** High-availability distributed storage clusters, snapshot management, and performance benchmarking.
- **HA Clustering:** Designing failover systems and fault-tolerant identity/compute clusters.
- **Linux (RHEL/Ubuntu):** Kernel tuning, package management, lights-out management (LOM), and shell automation.

DevOps, Security & Compliance

- **Ansible, SaltStack:** Automated configuration management, fleet audits, and rapid node provisioning.
- **CI/CD (GitHub Actions, Jenkins):** Automated test execution, artifact generation, and deployment pipelines.
- **FreeIPA, RBAC/HBAC:** Centralized authentication and role-based access control across thousands of heterogeneous nodes.

- **FedRAMP & STIG Compliance:** Automated compliance-as-code extraction, auditing, and continuous security baseline enforcement.
- **Qualys Security Intelligence:** Automated vulnerability scanning, compliance metric extraction, and remediation tracking.
- **Network Engineering & Firewalls:** Cisco routing/switching, IPsec VPNs, VLANs, Ruckus wireless bridges, and VoIP architecture.

Leadership, Mentoring & Methodology

- **Technical Mentorship:** Guiding multi-tier engineering teams, methodical troubleshooting, and stakeholder communication.
- **Robotics & STEM Mentoring:** Mentoring student teams in real-time embedded systems, electrical wiring, mechanisms, and troubleshooting.
- **Advanced Diagnostics & Monitoring:** Low-level system diagnostics, proactive performance telemetry, and elimination of systemic issues.
- **Rapid Adaptation / SME:** Proven track record of mastering unfamiliar architectures quickly and becoming an organizational SME.

PROFESSIONAL EXPERIENCE

Cisco Systems | Cloud Engineering Technical Leader

AUG 2019 - JULY 2026 (REMOTE)

Managed FedRAMP Moderate operations, backend data systems, and critical infrastructure for the Webex for Government platform, before transitioning in 2024 to the central SecOps engineering team to expand vulnerability management, automated tooling, and technical leadership across corporate Webex systems.

- **DevSecOps & Compliance Automation:** Developed automated reporting tools using the **Qualys API** to extract compliance metrics. Engineered Python scripts to parse raw **STIG XCCDF (XML)** data and map it to compliance results, enabling the automated generation of audit-ready STIG checklists. This compliance-as-code bridge eliminated hundreds of manual audit hours.
- **Data Platform Engineering:** Designed, deployed, and operated a vulnerability management platform using **Python (Pandas, SQLAlchemy)** and **PostgreSQL**. Built custom ETL pipelines to ingest and synchronize data from the **Jira API**, providing teams with high-fidelity dashboards for severity tracking and SLA compliance. Integrated data with **Tableau** to enable executive-level visibility into remediation progress.
- **Inventory Analytics & Infrastructure Operations:** Leveraged **Pandas** for large-scale data manipulation, merging disparate inventory reports and datasets into actionable operational insights across **Cisco UCS** bare-metal hardware and cloud compute. Transformed manual Excel-based tracking into automated trend analysis for monthly executive reporting.
- **Zero-Day Incident Response (Log4j):** Engineered and executed enterprise-wide **Ansible** playbooks during the 2021 Log4j vulnerability to perform deep filesystem audits across all running server fleets. Located dormant instances embedded in nested packages and non-standard directories that bypassed standard **Qualys** scans, auditing 100% remediation across the platform.
- **Technical Guidance & Mentorship:** Served as infrastructure team lead, providing technical direction across the Webex for Government environment. Mentored engineering teams in methodical troubleshooting and supported the analysis and resolution of complex vulnerability and compliance remediation issues, ensuring continuous service stability under strict SLAs.
- **Skills:** Python (Pandas, SQLAlchemy), PostgreSQL, Jira API, Qualys API, STIG Compliance (XCCDF/XML), ETL Pipelines, Tableau, Ansible, RHEL/Ubuntu, Cisco UCS, FedRAMP Moderate.

Cisco Systems | Cloud Systems Engineer

JAN 2017 - AUG 2019 (REMOTE)

Managed and optimized mission-critical compute infrastructure for Webex Commercial and FedRAMP environments. Collaborated with Networking and Storage teams to ensure zero-downtime availability, robust identity access control, and high security across US data centers.

- **Enterprise Identity Architecture:** Architected and deployed a **High-Availability (HA) FreeIPA cluster** to replace a legacy, high-overhead 389 Directory Server environment. Automated VM deployment in **OpenStack** using **Ansible**, enabling centralized RBAC and HBAC for 2,000+ Linux/Windows servers and infrastructure devices across multiple data centers.
- **Virtualized Infrastructure Operations:** Managed large-scale compute services using **vSphere** on **Cisco UCS** hardware. Administered critical infrastructure including bastion hosts, LDAP/FTP servers, and antivirus solutions, ensuring consistent high availability.
- **Vulnerability Management:** Led security patch remediation efforts for compute services in highly regulated federal environments. Researched and implemented system security patches, advised engineering teams on remediation strategies, and managed **Qualys** vulnerability scanners to maintain strict compliance.
- **Skills:** Ansible, FreeIPA, OpenStack, VMware vSphere, Cisco UCS, RHEL/CentOS, Windows Server, RBAC/HBAC, Qualys, FedRAMP Compliance.

Select Group @ Cisco | Cloud DevOps/QA Engineer (Contract)

FEB 2016 - JAN 2017 (REMOTE)

Provided specialized DevOps and QA engineering support for Cisco's OpenStack-based Public Cloud (InterCloud Services). Focused on performance optimization, automated configuration management, and stability engineering for large-scale cloud deployments.

- **Performance Optimization:** Identified and removed redundant **Nova Scheduler** filters, reducing instance scheduling time for 700 nodes from 20 seconds to 5 seconds—a **75% improvement** in orchestration speed.
- **Infrastructure-as-Code at Scale:** Engineered and executed **Ansible** playbooks to manage complex bare-metal configurations through **lights-out management (LOM)** interfaces and automated cron jobs across hundreds of production servers simultaneously.
- **Cloud Infrastructure Evaluation:** Stood up **RHUI (Red Hat Update Infrastructure)** and **Proxmox** clusters for **Proof-of-Concept (PoC)** assessments to determine viability for integration with Cisco InterCloud Services.
- **Stability Engineering:** Diagnosed and resolved a critical race condition involving cloudbase-init on Windows 2008/2012 images within **OpenStack**, eliminating intermittent reboots during initial instance setup.
- **Cloud QA & Orchestration:** Managed global test cycles for Compute and Heat components during major cloud upgrades. Developed **Python** and **Bash** scripts coupled with Ansible to perform comprehensive orchestration tests.
- **Skills:** OpenStack (Nova, Heat, Neutron, Glance), Ansible, Python, Bash, RHUI, Proxmox, Linux/Windows Administration, Performance Tuning, Cloud QA.

Australian Laboratory Services | Senior Systems & Storage Administrator

SEP 2014 - JAN 2016 (GRAND JUNCTION, CO)

Managed, modernized, and optimized enterprise storage, virtualization, database, and network infrastructure. Engineered high-availability distributed storage clusters, automated server provisioning, and implemented comprehensive monitoring to ensure maximum uptime, data integrity, and network performance.

- **Distributed Storage & Virtualization:** Engineered an Ubuntu-based **Ceph cluster** for **VMware ESXi** hosts, benchmarking iSCSI vs. RBD performance to optimize block storage access. Leveraged **Ceph RBD snapshots** to manage automated database schema testing, enabling a major production database upgrade with zero errors.
- **Infrastructure Automation:** Implemented **SaltStack** to manage Linux virtual environments and automate the configuration and rapid provisioning of new **Ceph** storage nodes and virtual machine resources.
- **Database & Data Integrity:** Wrote custom **Python (SQLAlchemy)** verification scripts to check for missing SQL foreign keys and ensure data integrity during major migrations. Configured replication and backups of production **MySQL** databases using **LVM snapshots** and **rsync**.
- **System Stability & Monitoring:** Installed **Observium** for performance telemetry and integrated proactive checks into **Nagios** to monitor production systems, decreasing service restarts from weekly to quarterly.
- **Network Hardware Engineering:** Troubleshoot and resolved a complex routing issue where **Asterisk** VoIP traffic was throttled by backup data tunnels; replaced overloaded gateway hardware, doubling available network bandwidth.
- **Skills:** Ubuntu/CentOS, VMware ESXi, Ceph RBD, iSCSI, SaltStack, Python (SQLAlchemy), MySQL, Nagios, Observium, Asterisk VoIP, Cisco/HP Routers & Switches, Sophos UTM, VLANs, IPsec VPN.

Garfield County School District 16 | Senior Systems/Network Administrator

OCT 2002 - JUN 2014 (PARACHUTE, CO)

Managed and modernized the district-wide enterprise network, systems, and core educational platform infrastructure. Led high-impact IT modernization projects, automated administrative processes, and engineered cost-effective systems that dramatically reduced operational costs and IT support overhead.

- **Systems Integration & Automation:** Developed a custom **Flask (SQLAlchemy)** web application that synchronized the centralized Student Information System (SIS) with **Active Directory** and **Google Apps**. Built self-service password reset and status tools for teachers, resulting in a **97% reduction in recurring support calls** to the helpdesk.
- **Enterprise Modernization & Development:** Spearheaded the replacement of a legacy, fragmented student information system with a web-based, centralized platform, eliminating manual data syncing between schools. Streamlined state compliance reporting by developing custom **PHP** generation scripts interfacing with the **PostgreSQL** database.
- **Infrastructure & Systems Virtualization:** Designed and administered virtualized host environments using **Citrix XenServer** and **VirtualBox** to run critical Windows Server 2008 and Ubuntu Linux servers. Supervised low-voltage cabling and hardware installation contractors for new facilities to ensure strict specification compliance.
- **Network Engineering:** Engineered a high-speed WAN connection between district buildings using **Ruckus wireless bridges**, increasing bandwidth by **6,500%** (1.5Mbps to 100+ Mbps) with zero ongoing monthly service costs. Managed core networking infrastructure including **Cisco Routers, Switches, PIX Firewalls**, and **Smoothwall** web filters.
- **Unified Communications & End-User Compute:** Deployed **Cisco CallManager** and **Cisco Unity** voice systems district-wide. Architected a centralized desktop computing solution using **Windows Terminal**

Servers and **HP Thin Clients** for 6 computer labs, saving the district over **\$60,000** in capital expenditure and simplifying end-user support.

- **Skills:** Python, Flask, SQLAlchemy, PHP, PostgreSQL, Ubuntu, Windows Terminal Server, Windows AD, Citrix XenServer, VirtualBox, Cisco CallManager/Unity, Cisco Routers & Switches, PIX, Ruckus Wireless Bridges/APs, HP Thin Clients, Google Apps, Smoothwall, Active Directory.

Foundational Experience | Systems Engineering, Software QA & Avionics Foundations

1991 - 2002 (VARIOUS)

Established a lifelong technical foundation by translating a low-level understanding of hardware, networking, and software into rigorous testing and validation methodologies. This ground-up engineering approach ensures comprehensive diagnostics and root-cause analysis at every layer of the stack.

- **Sales Engineering & Systems Integration:** Tested **ImageCast** software integrations with systems management suites including **SMS**, **CA Unicenter**, and **Desktop DNA**. Defined customer software requirements, submitted feature change requests, and conducted technical product demonstrations and training for enterprise clients such as **HP**, **Target**, and **NEC**. Helped answer technical questions at trade shows.
- **Technical Services & Test Lab Management:** Merged and managed QA testing with the technical support department. Designed test cases, executed testing plans, and managed the lab's **Windows** and **Linux** servers and test networks. Redesigned, built out, and managed the company's network, labs, and server infrastructure following a corporate split.
- **Software QA & Testing Leadership:** Promoted from contractor to QA Lead, managing all testing operations for **ImageCast** and **DrivePro** products. Structured software testing workflows by implementing the company's first bug-tracking system, drafting comprehensive test plans, and maintaining lab hardware and network infrastructure.
- **Training & Team Optimization:** Boosted support staff effectiveness and QA productivity by rotating tech support personnel through active lab testing scenarios. Trained technicians on networking topologies, testing methodologies, and product architecture to improve time-to-resolution for support issues.
- **Hardware Assembly & Diagnostics:** Built custom computers, performed component-level hardware and peripheral repairs, and conducted software troubleshooting. Developed quotes for client builds and mastered motherboard/component compatibility.
- **Airborne Radar & Electronics Engineering (U.S. Air Force):** Served as an Airborne Warning and Control Radar Journeyman (AWACS), repairing high-power microwave systems, high-voltage circuits, and analog/digital electronics.
- **Skills:** Analog & Digital Electronics, Circuit Analysis, System Diagnostics (Binary/Hex), PC Architecture, Disk Cloning (ImageCast, DrivePro), Software QA & Test Automation, Bug-Tracking Systems, Network Design, Windows/Linux Lab Administration, Technical Training, Requirements Engineering, Systems Management (SMS, CA Unicenter, Desktop DNA).

EDUCATION & CERTIFICATIONS

- **Linux Academy / A Cloud Guru:** AWS & Red Hat Enterprise Linux Curriculum
- **Regis University:** C Programming & Unix Systems Courses
- **Lincoln Christian University:** Music Theory & Composition
- **Colorado School of Mines:** 2 Years – Electrical Engineering & Applied Physics
- **Community College of the Air Force:** Avionics/Electronics Technician – RADAR Specialist (AWACS)

- **Field Training Detachment - Tinker AFB, OK:** AWACS RADAR and IFF Systems Journeyman – Awarded Senior Airman Below-the-Zone
- **Air Force Technical Training School - Keesler AFB, MS:** Electronics and AWACS RADAR Courses – Honor Graduate (99% Academic Test Average)

For a list of legacy certifications, see [linkedin.com/in/bjarrett/details/certifications/](https://www.linkedin.com/in/bjarrett/details/certifications/)